MODULE 05 · SECTION 1 - DESIGNING DATA PROCESSING SYSTEMS

Residency, Perimeters and Environments

Where data may live, where it may go, and how to stop a test from breaking production.

SECTION 1 of 5 EXAM WEIGHT 22% MODULE 5 of 45

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Location decides where data **rests**. A perimeter decides where it can **go**. Environments decide who can **break** it.

Region, dual-region, multi-region

LOCATION TYPE	EXAMPLE	WHAT IT PROMISES
`Region`	europe-west2, London	Data stays in that one region.
`Dual-region`	Two named regions, Cloud Storage	Two specific places, for resilience.
`Multi-region`	EU, US	Somewhere within a large area, not one country.

THE TRAP

If the rule names a country, a multi-region is not enough. EU keeps data in the EU, not in Germany.

BigQuery's location rules

- Location is set on the **dataset**, and **cannot be changed** after creation.
- A query **runs where its data lives**, and every table in it must share one location.
- A London table and a Frankfurt table **cannot be joined** in one query.
- To move data, **copy the dataset** to the new location, or recreate it there.

> ON THE EXAM
>
> A 'not found in location' error usually means the query mixed locations, not that the table is missing.

Residency or sovereignty?

DATA RESIDENCY

Where the data is

- Stored in a named region or country
- Processed there as well
- Set by locations and policies
- Easy to prove from configuration

DATA SOVEREIGNTY

Who can reach it

- Which laws and people can access it
- Support staff location and vetting
- Who holds the encryption keys
- Needs controls beyond location

THE TIEBREAKER

Residency is solved by location. Sovereignty adds people and keys: Assured Workloads, CMEK, Cloud EKM.

Layers that keep data in place

LAYER	WHAT IT DOES	COVERED IN
Dataset and bucket location	Puts the data in a region	Here
Processing location	Dataflow --region, BigQuery query location	Here, module 12
Resource locations constraint	Blocks creation anywhere else	Module 02
Assured Workloads	Locations, support staff and keys, as a package	Here
CMEK or Cloud EKM in region	Keys that also stay in place	Module 03

REMEMBER

A full residency design stacks these. One layer alone rarely satisfies a regulator.

Assured Workloads

- Applied to a **folder**. Every project inside inherits a **control package**.
- Packages are **enforced through organisation policies** and other features, so they cannot drift.
- Some packages limit **which Google support staff** can access your data, by location and vetting.
- Key options include **CMEK, Cloud EKM** and **Key Access Justifications**, depending on the package.

> ON THE EXAM
>
> 'A regulated workload, with personnel and location controls, least custom effort' points to Assured Workloads.

Assured Workloads packages, renamed June 2025

OLDER NAME	CURRENT NAME
EU Regions	EU Data Boundary
EU Regions and Support	EU Data Boundary and Support
Sovereign Controls for EU	EU Data Boundary with Access Justifications
UK Regions	UK Data Boundary

ON THE EXAM

Google says the functionality did not change. Recognise both names; answer with the question's.

IAM asks who you are. A perimeter asks where the request comes from, and where the data is going.

What VPC Service Controls stops

- **Copying out.** Data cannot be copied to resources outside the perimeter, even by someone allowed to read it.
- **Stolen credentials** used from networks outside the perimeter.
- **Insiders and compromised code** reaching services outside the perimeter.
- It does not replace IAM. Google recommends **both**, for defence in depth.

> HONEST LIMIT
>
> Some metadata, such as bucket names, is not perimeter-protected. Keep IAM tight there too.

Crossing the fence, on purpose

TOOL	WHAT IT DOES
Restricted services	The APIs the perimeter protects: BigQuery, Cloud Storage, Pub/Sub and more.
Access levels	Allow by context: source IP range, device, identity.
Ingress rule	Lets a named outside source in, to named resources.
Egress rule	Lets inside identities reach a named outside resource.
Perimeter bridge	Joins two perimeters both ways. Google prefers ingress and egress rules.
Dry-run mode	Logs violations without blocking anything.

THE TIEBREAKER

Bridges are bidirectional and coarse. For one partner or one flow, an ingress or egress rule is the answer.

Rolling out a perimeter without an outage

01

Dry run

Create the perimeter in dry-run mode. Nothing is blocked.

02

Read the logs

Every would-be violation is logged. Find the legitimate ones.

03

Add rules

Write ingress and egress rules for each legitimate flow.

04

Enforce

Switch to enforced mode once the logs are quiet.

ON THE EXAM

'Without disrupting existing pipelines' is a dry-run question.

IAM or VPC Service Controls?

IAM

Who may read it

- Grants a person or workload access
- Least privilege on datasets and buckets
- Cannot stop a reader copying it out
- Stolen credentials work anywhere

VPC SERVICE CONTROLS

Where it may go

- Fences projects and their services
- Blocks copies to outside resources
- Blocks valid credentials from outside
- Needs rules for legitimate flows

THE TIEBREAKER

'Even users who can read the data must not move it out', or 'exfiltration': VPC Service Controls.

Development and production, kept apart

- **Separate projects** for dev, test and prod, under separate folders with their own policies.
- **Separate service accounts**, so a dev pipeline physically cannot write to prod.
- **Promote through CI/CD**, never by hand. Module 18 builds that pipeline.
- **No real personal data in dev.** Use synthetic or de-identified samples, from module 04.

> THE PATTERN
>
> A project is the strongest cheap boundary you have: separate IAM, billing, quotas and perimeter.

What each level is good for

LEVEL	DRAW A BOUNDARY HERE FOR
Folder	Country, business unit or environment. Org policies and Assured Workloads.
Project	IAM, billing, quotas, VPC Service Controls perimeter membership.
Dataset	Location, default CMEK key, dataset access, time travel window.
Table and column	Row access policies, policy tags, masking. Module 29.

THE TAKEAWAY

Group data that shares the same rules. Larkspur's German data has its own folder, project and datasets.

Phrases that pick the boundary

THE QUESTION SAYS	REACH FOR
Data must stay **in one country**	Regional locations, plus the locations constraint
Regulated, with **support personnel** controls	Assured Workloads
Prevent **exfiltration**, or copying out	VPC Service Controls
One partner needs **access across** a perimeter	An ingress or egress rule
Without disrupting existing pipelines	Dry-run mode first
Dev must **not affect production**	Separate projects and service accounts

THE TIEBREAKER

Match the boundary to the risk: resting, moving, or breaking.

A Larkspur data scientist with read access to the curated dataset copied a customer table into a project outside Larkspur's warehouse projects. It was legitimate access, used to move data somewhere it should never go.

Security wants copying data out of the warehouse projects blocked, even for users who can read it, and even with stolen credentials.

What should security implement?

A Remove BigQuery Data Viewer from every data scientist

B Enforce domain restricted sharing across the organisation

C Put the warehouse projects in a VPC Service Controls perimeter

D Create a log-based alert that fires whenever a BigQuery copy job runs

◆ PAUSE AND DECIDE

What should security implement?

A **Remove Data Viewer from everyone**

Stops the copying by stopping the work. The scientists legitimately need to read the data.

B **Domain restricted sharing**

Controls who can be granted roles, not where data can be copied. The copier was already inside the domain.

C **A VPC Service Controls perimeter**

Copies to resources outside the fence are refused, whatever the reader's permissions, and stolen credentials fail from outside.

D **Alert on copy jobs**

Detects the copy after the data has left. The requirement was to block it.

THE SIGNAL

'Even users who can read it' and 'even with stolen credentials' is VPC Service Controls language.

> An Irish health-tech start-up is moving patient analytics to Google Cloud. Its regulator requires data to stay in the EU, and Google support staff who could access it to be EU-based.
>
> The start-up has a two-person platform team and wants the least custom configuration.

What should the start-up do?

A Use Assured Workloads with the EU Data Boundary and Support package

B Set the resource locations constraint to EU locations on the workload folder

C Create every BigQuery dataset and bucket in the EU multi-region

D Place all the workload's projects inside a VPC Service Controls perimeter

◆ PAUSE AND DECIDE

What should the start-up do?

A Assured Workloads, EU Data Boundary and Support

One package covers EU locations and EU-based support staff, enforced automatically.

B Resource locations constraint

Handles where resources are created, but says nothing about who at Google can support them.

C EU multi-region everywhere

Right place for the data, but relies on everyone remembering, and ignores the support staff rule.

D A VPC Service Controls perimeter

Stops data leaving through APIs. It does not set locations or control support personnel.

THE SIGNAL

Location plus personnel, least effort: Assured Workloads. Location alone would have been the constraint.

> An analyst joins German accounts in lk-de-warehouse-prod, located in europe-west3, with the tariffs table in lk-warehouse-prod, located in europe-west2. The query fails, saying a dataset was not found in location europe-west3.
>
> German personal data must not leave Germany. Tariffs contain prices only, with no personal data.

What should the team do?

A Rerun the query with its location set to the EU multi-region

B Copy the German accounts table into a London dataset, then join

C Grant the analyst BigQuery Job User on the German project

D Copy the tariffs table into a Frankfurt dataset, then join there

◆ PAUSE AND DECIDE

What should the team do?

A **Run it in the EU multi-region**

EU is a different location again. Neither regional dataset lives there, so the query still fails.

B **Copy German accounts to London**

It would work, and it breaks the rule. German personal data would leave Germany.

C **Grant Job User**

Not a permission problem. The error is about location, and the tables live in two places.

D **Copy tariffs to Frankfurt, join there**

Non-personal data moves to the personal data. The join works and nothing personal leaves Germany.

THE SIGNAL

'Not found in location' means mixed locations. Move the data that is allowed to move.

> Larkspur's pipeline developers test Dataflow jobs against the production BigQuery project. Last week a test deployment overwrote a production table.
>
> The team needs isolation, realistic test data, and no real customer data outside production.

What should the team change?

A Prefix test datasets with dev_ inside the production project, and agree a naming convention

B Separate dev and prod projects with their own service accounts, and give dev a de-identified sample

C Keep one project, and add IAM conditions so developers cannot write to tables starting with prod

D Copy all production data nightly into a dev project, so that tests are realistic

◆ PAUSE AND DECIDE

What should the team change?

A. A naming convention in prod

A convention is a promise, not a boundary. The next mistake writes to prod just as easily.

B. Separate projects, accounts, de-identified sample

A dev job cannot reach prod at all, and tests stay realistic without real customers.

C. IAM conditions on table names

Fragile: every new table needs the right name, and the pipeline's own account still writes to prod.

D. Copy all production data to dev

Realistic, isolated, and puts every customer's personal data into an environment built for experiments.

THE SIGNAL

Isolation is a project boundary. 'No real customer data' adds de-identification from module 04.

Module 05 in five lines

- Three boundaries: **where data rests**, **where it can go**, and **who can break it**.
- BigQuery location is **fixed per dataset**, and a query cannot **mix locations**.
- **Residency** is location. **Sovereignty** adds people and keys: Assured Workloads.
- **Exfiltration** means VPC Service Controls. Roll it out in **dry-run** first.
- **Dev and prod** are separate projects, with no real personal data in dev.

> COMING UP
>
> Next: module 06, reliability and fidelity: cleaning, validation and ACID.